

Amazon Web Services

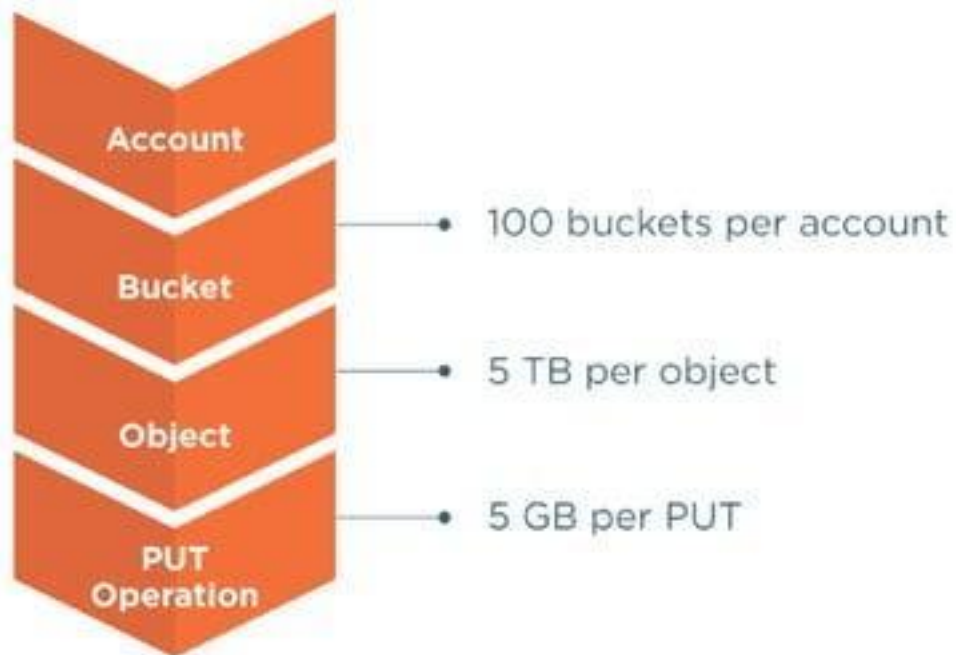
AWS S3: Simple Storage Service





- **S3 is a safe place to store your files.**
- **It is Object based storage.**
- **The data is spread across multiple devices and facilities**
- **S3 is Object based i.e. allows you to upload files.**
- **File size can be from 0 Bytes to 5 TB.**
- **There is unlimited storage**
- **Files are stored in Buckets.**
- **Account→Bucket -> Object**

S3 Maximums



- S3 bucket has a universal namespace, that is, names must be unique globally.
- <https://s3-eu-west-1.amazonaws.com/ashishrpandeyfirstbucket>
- When you upload a file to S3 you will receive a HTTP 200 code if the upload was successful.
- S3 provides developers and IT teams with secure, durable, highly-scalable object storage.
- Amazon S3 is easy to use, with a simple web services interface to store and retrieve any amount of data from anywhere on the web.

S3 – Buckets

- You can then upload any number of objects (pics, docs etc) to the bucket.
- An Amazon S3 bucket name is globally unique and buckets are created in a region you specify
- You can create up to 100 buckets in each of your AWS accounts.
- There is no limit to the number of objects that can be stored in a bucket and no difference in performance whether you use many buckets or just a few.
- You cannot create a bucket within another bucket.
- You can delete the bucket when not in use and that bucket name can be reused.
- If you want other users or AWS resources to access your bucket you have to make it public
- You can have multiple versions of objects in your bucket by enabling version control

S3 – Static Web hosting

- You can host a static website on Amazon S3
- Amazon S3 does not support server-side scripting.
- To host a static website, you configure an Amazon S3 bucket for website hosting, and then upload your website content to the bucket.
- This bucket must have public read access.
- The website is then available at the AWS Region-specific website endpoint of the bucket

S3 – Cross Region Replication

- Cross-region replication is a bucket-level configuration that enables automatic, asynchronous copying of objects across buckets in different AWS Regions
- These buckets can be owned by different AWS accounts.
- You need to add a replication configuration to your source bucket to direct Amazon S3 to replicate objects
- Amazon S3 stores object replicas using the same storage class as the source object

S3 Features

- Durability and availability
- Tiered Storage Available
- Lifecycle Management
- Versioning
- Encryption
- Secure your data using Access Control Lists and Bucket Policies

S3 Storage Classes:

- **S3-Standard** - 99.99% availability, 99.999999999% durability, stored redundantly across multiple devices in multiple facilities and is designed to sustain the loss of 2 facilities concurrently.
- **S3 - IA (Infrequently Accessed)**: For data that is accessed less frequently, but requires rapid access when needed. Lower fee than S3, but you are charged a retrieval fee.
- **S3-one-zone-IA** : For data that is accessed less frequently, but requires rapid access when needed, but replicated within one-zone only.
- **Reduced Redundancy Storage** - Designed to provide 99.99% durability and 99.99% availability of objects over a given year.
- **Glacier** - Very cheap, but used for archival only. It takes 3 – 5 hours to restore from Glacier.

S3 Storage Class Comparison:

	S3 Standard	S3 Standard-IA	S3 One Zone-IA	Amazon Glacier
Designed for Durability	99.999999999%	99.999999999%	99.999999999%†	99.999999999%
Designed for Availability	99.99%	99.9%	99.5%	N/A
Availability SLA	99.9%	99%	99%	N/A
Availability Zones	≥3	≥3	1	≥3
Minimum Capacity Charge per Object	N/A	128KB*	128KB*	N/A
Minimum Storage Duration Charge	N/A	30 days	30 days	90 days
Retrieval Fee	N/A	per GB retrieved	per GB retrieved	per GB retrieved**
First Byte Latency	milliseconds	milliseconds	milliseconds	select minutes or hours***
Storage Type	Object	Object	Object	Object
Lifecycle Transitions	Yes	Yes	Yes	Yes

S3 Storage Classes:

What is Glacier??

Glacier is an extremely low-cost storage service for data archival. Amazon Glacier stores data for as little as \$0.01 per gigabyte per month, and is optimized for data that is infrequently accessed and for which retrieval times of 3 to 5 hours are suitable.

S3 Charges:

Charged for

- Storage space
- Requests
- Storage Management Pricing
- Data Transfer Pricing
- Transfer Acceleration

For more details refer to :

<https://calculator.s3.amazonaws.com/index.html>

S3 Transfer Acceleration:

- Amazon S3 Transfer Acceleration enables fast, easy, and secure transfers of files over long distances between your end users and and S3 bucket.
- Transfer Acceleration takes advantage of Amazon CloudFront's globally distributed edge locations. As the data arrives at an edge location, data is routed to Amazon S3 over an optimized network path.

S3 Life Cycle Management:

- Can be used in conjunction with versioning.
- Can be applied to current versions and previous versions.
- Following actions can now be done
 - Transition to the Standard - Infrequent Access Storage Class (128Kb and 30 days after the creation date).
 - Archive to the Glacier Storage Class (30 days after IA, if relevant)
 - Expire

S3: Security and Encryption

S3 Securing your bucket:

- By default, all newly created buckets are PRIVATE
- You can setup access control to your buckets using;
 - Bucket Policies
 - Access Control Lists
- S3 buckets can be configured to create access logs which log all requests made to the S3 bucket. This can also be done to another bucket.

S3 Securing your bucket:

- In Transit
 - SSL/TLS

- At Rest

Server Side Encryption

- S3 Managed Keys – SSE(Server Side Encryption)-S3
- AWS Key Management Service, Managed Keys - SSE-KMS
- SSE with Customer Provided Keys -SSE-C

Client Side Encryption

S3 Server Access Logging

- To track requests for access to your bucket, you can enable access logging.
- Each access log record provides details about a single access request such as the requester, bucket name, request time, request action, response status, and error code
- Access log information can be useful in security and access audits.
- It can also help you learn about your customer base and understand your Amazon S3 bill.
- By default, logging is disabled. When logging is enabled, logs are saved to a bucket in the same AWS Region as the source bucket.